SOFTWARE REQUIREMENTS SPECIFICATION

for

Group No. 6 VaahanSetu

Version 1.0

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Revisions

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1 Introduction

VaahanSetu is a desktop Java application intended to support pre-trip bus allocation and crowd management and underutilization issue. It uses stored bus and route information together with passenger-demand estimates entered by a crowd management officer.

1.1 Document Purpose

This Software Requirements Specification, version 1.0, describes the requirements for VaahanSetu. It identifies the system's intended users, inputs, expected behavior, constraints, and outputs so that the team can design, implement, and test the same agreed scope. This draft covers the core allocation workflow and demand simulation.

1.2 Product Scope

VaahanSetu allows an administrator to select stored buses and routes for a planning session. A crowd management officer then enters expected passenger demand for each selected route and submits the values together. The application checks bus schedules, route duration, availability, and rest periods, then displays a proposed bus-to-route allocation. It compares the proposed allocation with the buses' original scheduled routes using the same demand values.

The system reports passengers accommodated, passengers left unaccommodated, and unused bus capacity for both scenarios. A simulation can change demand on multiple randomly selected routes every two minutes and trigger a recalculation.

1.3 Definitions, Acronyms and Abbreviations

Term	Definition
Allocation	Assignment of eligible buses to selected routes for a planning session.
Bus capacity	Maximum number of passengers a bus can carry, as stored in the bus record.
Crowd management officer	User who enters route demand and views allocation results.
Demand	Expected number of passengers needing transport on a selected route for the planning session.
Rest period	A scheduled time interval during which a bus must not be assigned to a route.
Unmet demand	Number of passengers who cannot be accommodated by the capacity assigned to a route or set of routes.
Unused bus capacity	Available passenger capacity assigned to routes that is not needed to accommodate the entered demand.

1.5 References and Acknowledgments

Project requirements and scope decisions provided by the VaahanSetu team. No external transport data source is assumed.

2 Overall Description

2.1 Product Overview

VaahanSetu is a self-contained desktop application for academic demonstration. It receives stored bus and route records and manual passenger-demand inputs, applies schedule and rest-period eligibility checks, and displays a proposed allocation and comparison results. The demand simulation is internal to the application and uses generated changes rather than live observations.

2.2 Product Functionality

- Authenticate the administrator and crowd management officer.
- Display stored bus records and route records.
- Allow the administrator to select buses and routes for a planning session.
- Allow the officer to enter and submit expected passenger demand for every selected route.
- Validate entered demand values.
- Check bus schedule, route duration, availability, and rest-period constraints.
- Produce and display proposed bus-to-route assignments.
- Compare original scheduled allocation and proposed allocation using the same demand values.
- Display passengers accommodated, unmet demand, and unused bus capacity for each scenario.
- Simulate demand increases and decreases on multiple randomly selected routes every two minutes.
- Recalculate and display assignments and comparison results after each simulated demand update.
- Optionally display route crowd levels using a heatmap or schematic visualization.

2.3 Design and Implementation Constraints

- The application shall be implemented as a desktop application in Java.
- The application shall use a database or equivalent persistent data store for sample bus, route, schedule, rest-period, and login records.
- The core allocation process shall use manually entered passenger-demand values and stored schedule information.
- A bus shall not be assigned to overlapping routes; it must be available for the full travel duration and outside any rest period.
- The simulation shall update demand every two minutes, with increases or decreases of 5–10 passengers on multiple randomly selected routes.
- Demand values shall not become negative.
- The allocation algorithm is not prescribed by this SRS; it must satisfy the specified inputs, constraints, and outputs.

2.4 Assumptions and Dependencies

- The project team supplies representative sample data for buses, routes, schedules, capacities, and rest periods.
- The entered demand values represent expected passenger demand before the buses begin their journeys, not live passenger counts at bus stops.
- The original-allocation comparison uses the buses' usual scheduled routes and the same demand inputs as the proposed allocation.
- The system can determine route duration and the relevant schedule/rest periods from stored records.
- Allocation and simulation results are required during the current session only; retaining them after the session is outside the current scope.
- A geographic heatmap depends on suitable route path or coordinate data and remains optional.

3 Specific Requirements

3.1 Functional Requirements

F1: The system shall provide a login interface for the administrator and crowd management officer and shall identify the user's role after successful authentication.

F2: The system shall display the stored bus records available for a planning session.

F3: The system shall display the stored route records available for a planning session.

F4: The system shall allow the administrator to select buses and routes for a crowd-management session.

F5: The system shall show the crowd management officer the buses and routes selected by the administrator.

F6: The system shall allow the officer to enter expected passenger demand for every selected route and submit the values together.

F7: The system shall validate demand inputs as valid non-negative passenger counts before processing an allocation.

F8: The system shall determine bus eligibility using the bus's schedule, route duration, availability, and scheduled rest periods.

F9: The system shall not assign a bus to a route if the route duration extends into the bus's next rest period or if the bus is otherwise unavailable for the full route duration.

F10: The system shall ensure that a bus is assigned to at most one route at a time within the proposed allocation.

F11: The system shall generate and display proposed bus-to-route assignments intended to reduce unmet passenger demand while respecting the defined bus-availability constraints.

F12: The system shall compare the original scheduled allocation and proposed allocation using the same entered passenger-demand values.

F13: For each comparison scenario, the system shall display passengers accommodated, passengers left unaccommodated, and unused bus capacity.

F14: After the officer starts the simulation, the system shall update demand every two minutes by increasing or decreasing demand by 5–10 passengers on multiple randomly selected routes, without allowing a route's demand to fall below zero.

F15: After each simulated demand update, the system shall recalculate and display revised bus-to-route assignments and updated comparison results, including unmet demand and unused bus capacity.

3.3 Use Case Model

The main actors are Administrator and Crowd Management Officer. The use-case model below represents the core functions. It is provided as PlantUML source so the team can render it as a UML use-case diagram using a compatible PlantUML tool.

```
@startuml
left to right direction
actor Administrator
actor "Crowd Management Officer" as Officer
rectangle VaahanSetu {
  usecase "U1: Log in" as U1
  usecase "U2: View buses and routes" as U2
  usecase "U3: Select buses and routes" as U3
  usecase "U4: Enter and submit demand" as U4
  usecase "U5: Calculate allocation" as U5
  usecase "U6: Compare allocations" as U6
  usecase "U7: Run demand simulation" as U7
  usecase "U8: View revised allocation and results" as U8
}
Administrator --> U1
Administrator --> U2
Administrator --> U3
Officer --> U1
Officer --> U4
Officer --> U8
U4 .> U5 : <<include>>
U5 .> U6 : <<include>>
U7 .> U5 : <<include>>
U7 .> U8 : <<include>>
@enduml
```

3.3.1 Use Case U1 — Log In

Author: Diksha Misra

Purpose: Authenticate a user and provide access according to the user's role.

Requirements Traceability: F1

Priority: High

Preconditions: The application is running and the user has a registered sample account.

Postconditions: The authenticated user can access the functions permitted for their role.

Actors: Administrator; Crowd Management Officer

Extends: None

Flow of Events

Basic Flow

1. The user opens the login screen.
2. The user enters credentials.
3. The system validates the credentials and identifies the user's role.
4. The system displays the appropriate role-specific interface.

Alternative Flow

5. If the credentials are invalid, the system displays an error and allows the user to try again.

Exceptions

6. If the login data store is unavailable, the system informs the user that authentication could not be completed.

Includes: None

3.3.2 Use Case U2 — View and Select Buses and Routes

Author: Pasupuleti Varshitha

Purpose: Allow the administrator to view available sample records and select the buses and routes for a planning session.

Requirements Traceability: F2, F3, F4, F5

Priority: High

Preconditions: The administrator is logged in and bus and route records are available.

Postconditions: The selected buses and routes are associated with the current planning session and are visible to the officer.

Actors: Administrator

Extends: None

Flow of Events

Basic Flow

7. The administrator opens the bus list.
8. The system displays available bus records.
9. The administrator opens the route list.
10. The system displays available route records.
11. The administrator selects the buses and routes for the session.
12. The system makes the selection available to the officer.

Alternative Flow

13. The administrator changes the selection before the officer submits demand.

Exceptions

14. If no records are available, the system displays an empty list and cannot proceed with a meaningful allocation until records are available.

Includes: None

3.3.3 Use Case U3 — Enter and Submit Passenger Demand

Author: Gurram Sri Amrutha

Purpose: Collect expected passenger demand for all selected routes and initiate the allocation process.

Requirements Traceability: F5, F6, F7

Priority: High

Preconditions: The officer is logged in and the administrator has selected buses and routes.

Postconditions: Valid demand values for all selected routes are submitted for allocation.

Actors: Crowd Management Officer

Extends: None

Flow of Events

Basic Flow

15. The officer views the selected routes.
16. The officer enters expected passenger demand for each route.
17. The officer submits all values together.
18. The system validates that each value is a valid non-negative passenger count.
19. The system begins allocation processing when all required values are valid.

Alternative Flow

20. If a value is missing or invalid, the system identifies the affected field and asks the officer to correct it.

Exceptions

21. If no buses or routes have been selected, the system informs the officer that the session is not ready for demand submission.

Includes: None

3.3.4 Use Case U4 — Calculate and Compare Bus Allocation

Author: Krishna Patidar

Purpose: Produce a proposed allocation subject to availability constraints and compare it with the original scheduled allocation.

Requirements Traceability: F8, F9, F10, F11, F12, F13

Priority: High

Preconditions: Valid demand values have been submitted and bus/route schedule information is available.

Postconditions: The system displays proposed assignments and comparison results for the original and proposed allocations.

Actors: Crowd Management Officer (initiates through submission); VaahanSetu system

Extends: None

Flow of Events

Basic Flow

22. The system checks bus schedules, route duration, availability, and rest periods.
23. The system excludes buses that cannot complete the route before their next rest period or are otherwise unavailable.
24. The system creates a proposed allocation without assigning a bus to overlapping routes.
25. The system evaluates the original scheduled allocation using the same demand values.
26. The system calculates passengers accommodated, unmet demand, and unused capacity for both scenarios.
27. The system displays the proposed bus-to-route assignments and comparison results.

Alternative Flow

28. If available capacity is insufficient for all demand, the system displays the allocation it can produce and the remaining unmet demand.

Exceptions

29. If schedule data required to evaluate a bus is missing or invalid, that bus cannot be reliably checked and should be excluded from eligible allocation pending correction.

Includes: None

3.3.5 Use Case U5 — Run Demand Simulation and Refresh Results

Author: Mrunal Sachin Patil

Purpose: Demonstrate how changes in expected demand affect bus allocation and crowd-management results.

Requirements Traceability: F14, F15

Priority: High

Preconditions: An initial allocation has been calculated and the officer starts the simulation.

Postconditions: The displayed demand, proposed bus assignments, unmet demand, and unused capacity reflect the latest simulated demand update.

Actors: Crowd Management Officer; VaahanSetu system

Extends: None

Flow of Events

Basic Flow

30. The officer starts the simulation.
31. The system waits for the two-minute update interval.

32. The system randomly selects multiple routes and increases or decreases each selected route's demand by 5–10 passengers.
33. The system prevents any demand value from falling below zero.
34. The system recalculates the proposed allocation and comparison statistics.
35. The system displays the revised assignments, demand values, unmet demand, and unused capacity.
36. The system repeats the update cycle while the simulation remains active.

Alternative Flow

37. A route's demand may remain unchanged during an update if it is not selected.

Exceptions

38. If a recalculation cannot be completed because required schedule data is invalid, the system should display an error rather than present the failed calculation as a valid result.

Includes: None

4 Other Non-functional Requirements

4.1 Performance Requirements

- NFR-P1: For the team's sample dataset, the system should display the initial allocation and comparison results promptly after valid demand is submitted.
- NFR-P2: Each simulation update shall occur at the defined two-minute interval while the simulation is active.
- NFR-P3: After a simulation update, revised assignments and comparison statistics should be displayed before the next update interval.
- NFR-P4: The system should remain responsive while displaying the selected bus/route records and allocation results.

4.2 Safety and Security Requirements

- NFR-S1: The system shall require login before exposing role-specific functions.
- NFR-S2: The system shall restrict bus/route selection functions to the administrator and demand-entry/result functions to the crowd management officer, according to the defined roles.
- NFR-S3: The system shall validate passenger demand as a non-negative count and shall prevent simulated demand from becoming negative.
- NFR-S4: The system shall not present simulated passenger-demand changes as actual live passenger data.
- NFR-S5: Invalid or incomplete schedule information shall not be silently treated as proof that a bus is available.

5 Other Requirements

The route heatmap is an optional extension. If implemented, it may show the selected routes together and use route colors to represent crowding relative to the capacity allocated to each route. A geographic representation depends on suitable route path or coordinate data. If such data is not available, the core application shall still function without a geographic heatmap.

The system is intended for a single planning session at a time for this project draft. Demand inputs and allocation results need not be retained after the session ends.

Appendix A — Activity Log

To be completed by the team lead in accordance with the course template. Record meeting dates, team members present, work completed, decisions made, and planned next steps.

Date	Members Present	Activity / Decision	Next Steps
11-10-2026	Pasupuleti Varshitha Diksha Misra Gurram Amrutha Sri Krishna Patidar Mrunal Sachin Patil	Workflow and distribution of responsibilities.	Execution.